The negative sign implies that both are in opposite directions. The above equation can be written as

$$\frac{d^2\theta}{dt^2} = -\frac{p_m B}{I}\sin\theta$$

This is non-linear second order homogeneous differential equation. In order to make it linear, we use small angle approximation as we did in XI volume II (Unit 10 – oscillations, Refer section 10.4.4) i.e., $\sin\theta \approx \theta$, we get

$$\frac{d^2\theta}{dt^2} = -\frac{p_m B}{I}\theta$$

This linear second order homogeneous differential equation is a Simple Harmonic differential equation. Therefore,

Comparing with Simple Harmonic Motion (SHM) differential equation

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

where ω is the angular frequency of the oscillation.

$$\omega^2 = \frac{p_m B}{I} \Rightarrow \omega = \sqrt{\frac{p_m B}{I}}$$

$$T = 2\pi\sqrt{\frac{I}{p_m B}}$$

$$T = 2\pi\sqrt{\frac{I}{p_m B_H}} \text{ in second}$$

where, B_H is the horizontal component of Earth's magnetic field.

3.3.1. Potential energy of a bar magnet in a uniform magnetic field

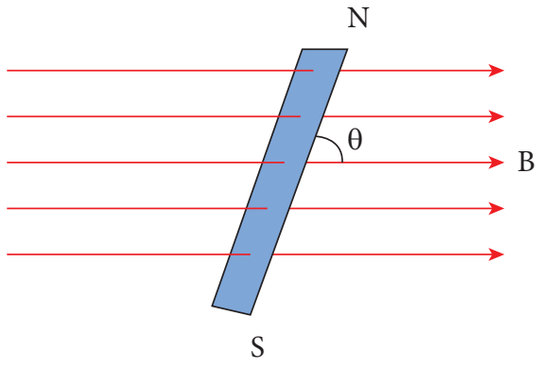


Figure 3.20: A bar magnet (magnetic dipole) in a uniform magnetic field

When a bar magnet (magnetic dipole) of dipole moment $\vec{p}_m$ is held at an angle θ with the direction of a uniform magnetic field $\vec{B}$, as shown in Figure 3.20 the magnitude of the torque acting on the dipole is

$$|\vec{\tau}_B| = |\vec{p}_m||\vec{B}|\sin\theta$$

If the dipole is rotated through a very small angular displacement dθ against the torque τ_B at constant angular velocity, then the work done by external torque $(\vec{\tau}_{ext})$ for this small angular displacement is given by

$$dW = |\vec{\tau}_{ext}|d\theta$$

The bar magnet has to be moved at constant angular velocity, which implies that $|\vec{\tau}_B| = |\vec{\tau}_{ext}|$

$$dW = p_m B\sin\theta\, d\theta$$

Total work done in rotating the dipole from θ′ to θ is

$$W = \int_{\theta'}^{\theta} \tau d\theta = \int_{\theta'}^{\theta} p_m B\sin\theta\, d\theta = p_m B\left[-\cos\theta\, d\theta\right]_{\theta'}^{\theta}$$

$$W = -p_m B(\cos\theta - \cos\theta')$$

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